CO3 - AT2

DATA INTERPRETATION

Sampling Distributions, Covariance, Correlation, Causation & Missing Data

SYED MUHAFIZ A

Register No.: 192472282

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Saveetha School of Engineering (SIMATS)

1. Standardization / Normalization

Problem: "Age" ranges from 18–75 and "Income" ranges from ₹20,000–₹20,00,000 in a dataset meant for a distance-based algorithm (e.g. KNN, K-Means, SVM with RBF kernel). Explain why failing to standardize distorts analysis, and calculate the z-score for Age = 45 given mean = 40, SD = 12.

Why unstandardized features distort distance-based analysis

Distance-based algorithms compute similarity using metrics such as Euclidean distance:

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d(x, y) = √[ (Age_x - Age_y)² + (Income_x - Income_y)² ]
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The core issue is scale dominance. A typical difference in Age between two people might be 10–20 units, while a typical difference in Income could be 5,00,000+ units. When these raw values are squared and summed, the Income term will numerically overwhelm the Age term — not because income is more important, but purely because its numeric scale is roughly 50,000 times larger.

- Reasoning, not just magnitude: The algorithm has no concept of "units" — it only sees numbers. It cannot tell that ₹20,000 and 20 years are conceptually different scales; it treats a difference of 20,000 in Income as "bigger" than a difference of 57 in Age, even if, in real-world terms, the age gap is far more decisive for the classification or clustering task.
- Effective feature deletion: Because Income dominates the distance formula, Age's contribution shrinks to near-zero. The model effectively behaves as if it were clustering/classifying on Income alone — Age becomes statistical dead weight despite being potentially just as predictive.
- Gradient-based learners are affected too: Even though the question specifies distance-based methods, it is worth noting that gradient descent-based models (e.g., logistic regression, neural nets) also converge poorly on unscaled features because the loss surface becomes elongated and ill-conditioned along the large-scale axis.

Standardization (z-score scaling) removes this artificial scale bias by re-expressing every feature in terms of "number of standard deviations from its own mean." After transformation, both Age and Income live on the same dimensionless scale, so distance calculations reflect genuine relative variation in each feature rather than the arbitrary units they happened to be recorded in.

Z-score calculation for Age = 45

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Formula:  z = (x - μ) / σ
Given:    x = 45,    μ (mean) = 40,    σ (SD) = 12
z = (45 - 40) / 12
z = 5 / 12
z ≈ 0.4167
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Answer: z ≈ 0.42

Interpretation: An age of 45 lies about 0.42 standard deviations above the mean age of 40. This is a mild, unremarkable deviation — nowhere near outlier territory (which conventionally starts around $|z| > 3$) — meaning 45 is a fairly typical value within this distribution of ages.

2. Distribution Shape from Summary Stats Alone

Problem: Mean = 45, Median = 45, Mode = 45, skewness ≈ 0, but kurtosis = 5.8 (leptokurtic). What does this combination reveal that skewness and central tendency alone would miss?

What the equal Mean = Median = Mode and zero skewness tell us

When mean, median and mode coincide and skewness is approximately zero, the standard conclusion is that the distribution is symmetric — there is no long tail pulling the mean away from the median, so left and right sides mirror each other. Taken alone, an analyst might reasonably (but incorrectly) assume this means the data resembles a Normal distribution.

What kurtosis = 5.8 adds — the missing dimension

Kurtosis measures the shape of the tails and the peak, independent of central tendency and independent of symmetry. A Normal distribution has kurtosis = 3 (mesokurtic). A value of 5.8 is markedly higher, so the distribution is leptokurtic. This tells us something that symmetry and central-tendency measures are structurally unable to reveal:

a. Sharper, taller peak: Values are more tightly clustered around the mean than in a Normal distribution — the "typical" observation is even closer to 45 than a Normal curve with the same variance would suggest.

b. Fatter tails (more extreme values): Simultaneously, there is a higher probability mass in the extreme tails. Leptokurtic distributions produce more frequent large deviations (both positive and negative) than a Normal distribution — informally, "more outliers, more often."

c. Thinner shoulders: The probability mass is redistributed away from the moderate-deviation region (the "shoulders") into both the center and the extreme tails, giving the density curve a narrow, peaked look with long, thin tails — sometimes visualized as more like a Laplace/double-exponential shape than a bell curve.

d. Symmetric ≠ Normal: Crucially, because skewness ≈ 0, the extra tail risk is balanced — it isn't a one-sided risk. But mean/median/mode and skewness only rule out asymmetry; they say nothing about tail thickness. Two distributions can have identical mean, median, mode, and skewness of 0, yet one could be perfectly Normal (kurtosis 3) and the other heavy-tailed (kurtosis 5.8) — central tendency and skewness simply cannot distinguish between them.

Practical implication: If this were, say, a distribution of daily investment returns or process measurements, relying only on mean/median/mode/skewness would create false confidence that the data is "well-behaved" like a Normal distribution. In reality, the high kurtosis warns that extreme events (large gains/losses, defects, spikes) occur more often than a Normal-based model (e.g. one using $\pm 3\sigma$ control limits or Normal-based confidence intervals) would predict. Risk models, control charts, and hypothesis tests that assume normality would understate the true probability of extreme outcomes.

3. Grouped Data — Mean and Variance (Step-Deviation Method)

Data given:

Class Interval	0–10	10–20	20–30	30–40	40–50
Frequency (f)	5	8	15	12	10

Step 1: Assumed mean and midpoints

We take class width h = 10, and assume the mean A at the midpoint of the class with the highest frequency (20–30), so A = 25. Midpoints of each class:

Midpoints: 5, 15, 25, 35, 45 (i.e. midpoint of each 10-unit interval)

Step 2: Compute d = (x − A) / h and the working table

Class	Midpoint (x)	f	d = (x–A)/h	f·d	d^2	$f \cdot d^2$
0–10	5	5	−2	−10	4	20
10–20	15	8	−1	−8	1	8
20–30	25	15	0	0	0	0
30–40	35	12	1	12	1	12
40–50	45	10	2	20	4	40
Total	—	N = 50	—	Σfd = 14	—	$\Sigma fd^2 = 80$

Step 3: Mean by step-deviation method

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Mean = A + (Σfd / N) × h
     = 25 + (14 / 50) × 10
     = 25 + 2.8
     = 27.8
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Mean = 27.8

Step 4: Variance by step-deviation method

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Variance = h² × [ (Σfd² / N) − (Σfd / N)² ]
         = 10² × [ (80/50) − (14/50)² ]
         = 100 × [ 1.6 − (0.28)² ]
         = 100 × [ 1.6 − 0.0784 ]
         = 100 × 1.5216
         = 152.16
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Variance = 152.16 (Standard Deviation = √152.16 ≈ 12.34)

4. Comparing Two Distributions via Coefficient of Variation

Given: Portfolio A — Mean return = 12%, SD = 3%. Portfolio B — Mean return = 18%, SD = 5%.

Claim to evaluate: "Portfolio B is riskier in absolute terms but not necessarily worse per unit of return."

What CV captures that SD alone doesn't

Standard deviation is an absolute measure of dispersion — it is expressed in the same units as the variable itself (here, percentage return points). This makes it unsuitable for comparing risk across two distributions that have different means or different scales, because a larger SD might simply reflect a larger average value rather than genuinely greater relative volatility. The Coefficient of Variation normalizes SD by the mean, converting risk into a relative, unit-free measure — "risk per unit of expected return":

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CV = (SD / Mean) × 100%
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Calculation

Portfolio	Mean Return	SD	CV = (SD/Mean)×100
A	12%	3%	(3/12)×100 = 25.00%
B	18%	5%	(5/18)×100 = 27.78%

Evaluating the claim

The first half of the claim is correct: Portfolio B does carry more absolute risk — its SD of 5% is numerically larger than A's 3%, meaning B's returns swing further in raw percentage-point terms.

The second half of the claim does not hold up quantitatively. Portfolio B's CV (≈27.78%) is higher than Portfolio A's CV (25.00%). This means that for every 1% of expected return, Portfolio B carries more risk than Portfolio A does — B is not just riskier in absolute terms, it is also riskier per unit of return. So contrary to the investor's claim, B is the relatively less efficient (worse risk-adjusted) option of the two, despite its higher headline return.

- A naive comparison using SD alone might wrongly favor Portfolio A simply because 3% < 5%, without accounting for the fact that A also earns a smaller return, so some extra absolute risk in B could theoretically be "worth it."
- The CV comparison settles this precisely: because 27.78% > 25.00%, B's extra return (18% vs 12%) does not fully compensate for its extra risk — an investor is taking on proportionally more uncertainty for each percentage point of return earned in Portfolio B.
- This is exactly the scenario CV is designed for: comparing risk across assets/distributions with different means, which raw SD cannot do meaningfully on its own.

Conclusion: CV(A) = 25.00% < CV(B) = 27.78% → Portfolio A is more risk-efficient. The claim is only half correct — B is riskier both in absolute terms and per unit of return.

5. Outlier's Effect on Variance

Given: n = 10 values, original variance = 25. One value, originally 40 (close to the mean of 42), is mistakenly recorded as 400. Explain directionally the effect on (a) the mean and (b) the variance, and why variance is affected disproportionately more.

(a) Effect on the mean

The recording error adds an excess of (400 − 40) = 360 to the sum of the dataset. Since the mean is the sum divided by n, this excess is diluted across all 10 observations:

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Δ(Mean) = Δ(Sum) / n = 360 / 10 = 36
New mean ≈ 42 + 36 = 78  (directionally — a large but linear, damped shift)
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Direction: The mean increases substantially (from ~42 to roughly 78), but the error's impact is dampened by a factor of n — every one of the 10 observations "absorbs" only one-tenth of the erroneous excess. The mean shifts, but only linearly and proportionally to 1/n.

(b) Effect on the variance

Variance is built from squared deviations from the mean:

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Variance = (1/n) Σ (xᵢ - mean)²   [or divided by (n-1) for sample variance]
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Two compounding effects hit the variance simultaneously:

- The erroneous point's own deviation explodes: Before the error, 40 was close to the mean (42), contributing a small squared deviation of about $(40-42)^2 = 4$. After the error, even using the shifted mean of ~78, the deviation becomes $(400-78)^2 \approx 322^2 \approx 103{,}684$ — a squared deviation roughly 25,000 times larger than before.
- Every other point's deviation also grows: Because the mean itself has jumped from 42 to ~78, all nine of the other, unaffected values are now measured against a mean that has shifted by 36. Each of their squared deviations increases too (a value near the old mean of 42 now deviates from the new mean of 78 by about 36 more than before), inflating the sum of squares even for "clean" data points.
- Squaring amplifies extremity nonlinearly: A 10× error in one raw value becomes roughly a 100× effect on that value's contribution to the sum of squares, because variance uses squared, not linear, deviations. This is the core mathematical reason variance (and SD) are so sensitive to outliers — the squaring operation deliberately/structurally penalizes large deviations far more than proportionally.

Direction and magnitude: The variance will increase dramatically — not by a modest multiple, but potentially by one or two orders of magnitude (from 25 to a figure plausibly in the thousands), even though only 1 of 10 data points was corrupted.

Why variance is disproportionately more affected than the mean

- Linear vs. quadratic sensitivity: The mean is a linear function of the data (it responds to errors proportionally, damped by 1/n). Variance is a quadratic function of the data (deviations are squared before averaging), so the same raw error of +360 contributes to the mean only through its magnitude, but contributes to the variance through its magnitude squared.

- No self-cancellation for variance: If the erroneous value had been, say, symmetric around the mean (e.g., an equally-sized error in the opposite direction on another point), the two errors could partially cancel out in the mean. But squared deviations are always positive, so every large deviation — regardless of its sign — only ever adds to the variance; there is no cancellation effect to dampen it the way there can be for the mean.
- Leverage effect: A single extreme point acts like a lever arm — the further it sits from the mean, the more disproportionate its (squared) contribution to total dispersion. This is precisely why variance, SD, and any squared-error-based statistic (e.g. R^2, MSE) are described as "sensitive to outliers," while median-based or rank-based statistics are comparatively robust.

Summary: The mean rises noticeably but proportionally (damped by 1/n, ≈42→78). The variance rises far more sharply and disproportionately, because it depends on the squared deviation of the corrupted point (which itself has ballooned) and because the shift in the mean inflates every other point's squared deviation as well — a compounding, quadratic effect that a single linear-scale error cannot produce in the mean.